

CS5275 The Algorithm Designer's Toolkit
S2 AY2025/26
Midterm exam

6:30 pm – 8:15 pm, March 6 (Friday)

Note: All graphs are simple (no self-loops and multi-edges) and undirected.

True or false (14 marks: 1 mark per question)

Which of the following statements are correct?

For each of the statements, answer **true** (T or $\bigcirc$) or **false** (F or $\times$).

1. Let $G = (V, E)$. If we let H to be a graph over the same vertex set V with conductance $\Theta(1)$ and maximum degree $\Theta(1)$, then $\Phi(G, H)$ is within a constant factor of the edge expansion of G .
2. For any cycle graph, it is possible to remove at most 0.01 fraction of the edges so that each remaining connected component has conductance $\Omega(1)$.
3. For any connected graph G , if the largest eigenvalue of the normalized Laplacian is at least $2(1 - \epsilon)$, then G is $O(\epsilon)$ -close to bipartite.

Hint: The following variational characterization for general graphs might be useful.

$$\lambda_n = 2 - \min_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{\sum_{v \in V} \deg(v) x_v^2}$$

4. $\Phi(G) \in \Omega(1)$ implies $T_{\text{mix}}(G, 0.001) \in O(1)$.
5. The transition matrix of the standard (non-lazy) random walk is symmetric even if the graph is not regular.
6. For $d \geq 3$, the diameter of any n -vertex d -regular Ramanujan graph is $O(\log n)$.
7. Let $G = (V, E)$ be any n -vertex graph. Independently sample k unordered vertex pairs

$$\{s_1, t_1\}, \{s_2, t_2\}, \dots, \{s_k, t_k\},$$

where each pair is chosen uniformly at random from the $\binom{n}{2}$ unordered pairs of distinct vertices.

There exists a choice of parameter $k \in O(n \log n)$ such that, with probability at least $1 - \frac{1}{n}$, there exists an index i such that

$$\kappa_G(s_i, t_i) = \kappa(G),$$

where $\kappa(G)$ denotes the global vertex connectivity of G .

8. Let $G = (V, E)$ be any graph whose vertex connectivity is $\kappa(G) = k$. Let $S \subseteq V$ be an arbitrary set of k vertices. Construct a graph G' by adding a new vertex r that is adjacent to every vertex in S . Then we must have $\kappa(G') = k$.

Let α be any real number such that $0 < \alpha < 1$. Let $G = (V, E)$ be any connected d -regular graph, let $\mathbf{A}$ be its adjacency matrix, and define

$$\mathbf{W} = \frac{1}{d}\mathbf{A}$$

to be the transition matrix of the standard random walk. Define the α -lazy walk by

$$\mathbf{W}_\alpha = \alpha I + (1 - \alpha)\mathbf{W}.$$

9. The uniform distribution is a stationary distribution for the α -lazy walk.
 10. The α -lazy walk converges to a stationary distribution.

In the class, when we discuss the power method, we restrict our attention to the case where all the eigenvalues are non-negative. Now we revisit the power method again, but without this assumption.

Let $\mathbf{M} \in \mathbb{R}^{n \times n}$ be any real symmetric matrix, and let $\mathbf{x}_0 \in \mathbb{R}^n$ be any vector such that $\mathbf{M}\mathbf{x}_0 \neq \mathbf{0}$. Consider the vectors resulting from the power method:

$$\mathbf{x}_{t+1} = \mathbf{M}\mathbf{x}_t, \quad t = 0, 1, 2, \dots$$

Define the Rayleigh quotient at time t by

$$\rho_t = \frac{\mathbf{x}_t^\top \mathbf{M} \mathbf{x}_t}{\mathbf{x}_t^\top \mathbf{x}_t}.$$

Define the normalized vector at time t by

$$\mathbf{y}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$$

11. The sequence of Rayleigh quotients $\{\rho_t\}_{t \geq 0}$ always converges.
 12. The sequence of normalized vectors $\{\mathbf{y}_t\}_{t \geq 0}$ always converges.

We say that a graph $G = (V, E)$ is C -robust against edge deletion if the following condition is satisfied for any edge set $D \subseteq E$ (we allow $D = \emptyset$):

$$\sum_{\substack{S \text{ is a small connected component of } G'}} \text{vol}_G(S) \leq C \cdot |D|,$$

where $G' = (V, E \setminus D)$ is the graph resulting from removing D from G , and we say that a connected component $S \subseteq V$ of G' is small if $\text{vol}_G(S) \leq \text{vol}_G(V)/2$.

13. If $\Phi(G) \geq \phi$, then G is $O(1/\phi)$ -robust against edge deletion.
 14. If G is C -robust against edge deletion, then $\Phi(G) \in \Omega(1/C)$.

Multiple choice questions (6 marks: 1 mark per question)

For each of the following questions, **select the correct answer** from the five choices.
Each question has exactly one correct answer.

1. What is the conductance of an n -clique?

- (a) 1
- (b) $1 - \Theta(1/n)$
- (c) $\frac{1}{2} + \Theta(1/n)$
- (d) $\frac{1}{2} - \Theta(1/n)$
- (e) None of the above

In **Questions 2–3**, we consider Bourgain’s embedding with a single sampling probability from **Problem Set 1**. Let $G = (V, E)$ be a connected unweighted graph with shortest-path metric dist_G . Sample a random subset $A \subseteq V$ by including each vertex independently with probability p , and define the embedding into the line:

$$f_A(x) := \text{dist}_G(x, A) := \min_{a \in A} \text{dist}_G(x, a),$$

which yields the following line metric:

$$d_A(x, y) = \begin{cases} |f_A(x) - f_A(y)|, & \text{if } A \neq \emptyset; \\ 0, & \text{if } A = \emptyset. \end{cases}$$

2. Suppose $G = (V, E)$ is an n -vertex path.

Which choice of sampling probability p ensures that

$$\forall x, y \in V, \quad \mathbb{E}[d_A(x, y)] \in \Theta(1) \cdot \text{dist}_G(x, y) ?$$

- (a) $1/2$
- (b) $1/\log n$
- (c) $1/\sqrt{n}$
- (d) $1/n$
- (e) None of the above

3. Suppose $G = (V, E)$ is an n -vertex clique.

Which choice of sampling probability p ensures that

$$\forall x, y \in V, \quad \mathbb{E}[d_A(x, y)] \in \Theta(1) \cdot \text{dist}_G(x, y) ?$$

- (a) $1/2$
- (b) $1/\log n$
- (c) $1/\sqrt{n}$
- (d) $1/n$
- (e) None of the above

In **Questions 4–6**, let $G = (V, E)$ be an n -vertex tree, let $\mathbf{A}$ be its adjacency matrix, and let $\lambda_1, \dots, \lambda_n$ denote the eigenvalues of $\mathbf{A}$.

Hint: The trace of a matrix, which is the sum of the elements on its main diagonal, equals the sum of all its eigenvalues.

4. What is the value of $\sum_{i=1}^n \lambda_i$?

- (a) $2(n-1)$
- (b) $n-1$
- (c) 1
- (d) 0
- (e) None of the above

5. What is the value of $\sum_{i=1}^n \lambda_i^2$?

- (a) $2(n-1)$
- (b) $n-1$
- (c) 1
- (d) 0
- (e) None of the above

6. What is the value of $\sum_{i=1}^n \lambda_i^3$?

- (a) $2(n-1)$
- (b) $n-1$
- (c) 1
- (d) 0
- (e) None of the above

Proof-based questions (10 marks: 3+3+4)

1. Let G be any n -vertex d -regular graph, and let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of its normalized Laplacian $\mathbf{N}$.
 - (a) Prove that if G has at least k connected bipartite components, then $\lambda_{n-k+1} = 2$.
 - (b) Prove that if $\lambda_{n-k+1} = 2$, then G has at least k connected bipartite components.
2. Prove that, if $n \geq 100 \cdot d$, then any clique in any n -vertex d -regular Ramanujan graph has at most $O(\sqrt{d})$ vertices.

Grading: Problems 1(a) and 1(b) are worth 3 marks each, and Problem 2 is worth 4 marks.

Answer sheet

Student name:

Student ID:

True of false	Answer
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

Multiple choice questions	Answer
1	
2	
3	
4	
5	
6	

For the proof-based questions, write your solutions in any available space on this paper or on additional blank sheets.